

Non-Linear Budget Constraints

Seven subsidy schemes bend the budget line in different places. The question is the same in every one: where does the best affordable bundle land, and does it land on the fold?

Non-Linear Budget Constraints

In-kind subsidies, partial subsidies and exclusive schemes. Each one bends the budget line somewhere different, and the optimum can land right on the fold.

ES EN Oscuro Claro

SCHEME

- 0 Simple constraint
- 1 In-kind subsidy (non-exclusive)**
- 2 Partial in-kind subsidy (non-exclusive)
- 3 Exclusive subsidy (public schools)
- 4 Partial exclusive subsidy
- 5 Cash equivalent (full)
- 6 Cash equivalent (partial)

COMPARE WITH

Overlays a second scheme. The comparison that matters is against the cash transfer of equal cost.

- None
- 0 Simple constraint
- 1 In-kind subsidy (non-exclusive)
- 2 Partial in-kind subsidy (non-exclusive)
- 3 Exclusive subsidy (public schools)
- 4 Partial exclusive subsidy
- 5 Cash equivalent (full)**
- 6 Cash equivalent (partial)

PREFERENCES

- Cobb-Douglas** Perfect substitutes
- Perfect complements Quasilinear
- Generalised CES X inferior
- Custom

$$u(x, y) = x^a \cdot y^{(1-a)}$$

a

PRICES AND INCOME

p_x
 p_y
 m

THE PROGRAMME

x_s
 The allowance x_s is handed over free.

VIEW

- ☒ Heat gradient
- Turn it off and the background goes plain. Indifference curves keep their height colour at all times.
- ☒ Substitution panel
- $x \leq$ $y \leq$
- ☐ Fit the axes automatically

LAYERS

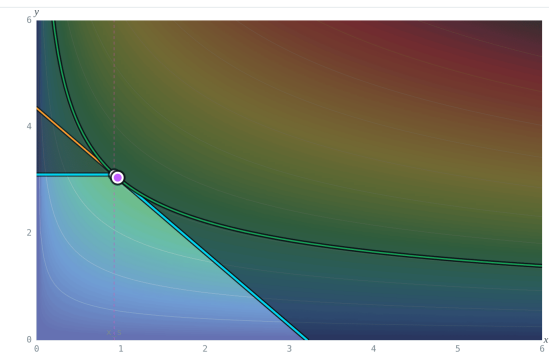
- ☒ Background contours
- ☒ Feasible set
- ☒ Curve through the optimum
- ☒ Optimum
- ☒ Vertices
- ☒ Compared set
- ☒ Grid

Cash and kind tie

The in-kind optimum falls on the stretch where the frontier coincides with the cash one, so the allowance does not bind: the two transfers deliver exactly the same.

PLANE (X, Y)

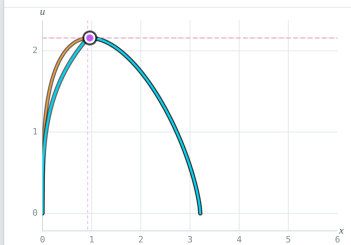
Scheme Compared Optimum Indifference



x^*	y^*	$U(x^*, y^*)$	COST
0.96	3.05	2.158	3.50

UTILITY ALONG THE CONSTRAINT

Scheme Compared



KIND OF OPTIMUM	CASH ADVANTAGE	JUMP AT x_s
Tangency	0.000	0.00

Translated from a GeoGebra applet. Syntax: + - * / ^ () , sqrt ln log exp abs min max sin cos, constants e and pi. Implicit multiplication works: 2xy = 2*x*y.

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1 What this is

A textbook budget constraint is a straight line. Almost no public programme leaves one straight. A quota of free milk, a discount on the first few units, a place at a state school that is forfeited the moment you buy a private one: each bends the frontier somewhere different, and the optimum can land exactly on the fold.

This applet carries seven schemes, the ability to compare two at once, and an optimum computed in a way that actually finds folds.

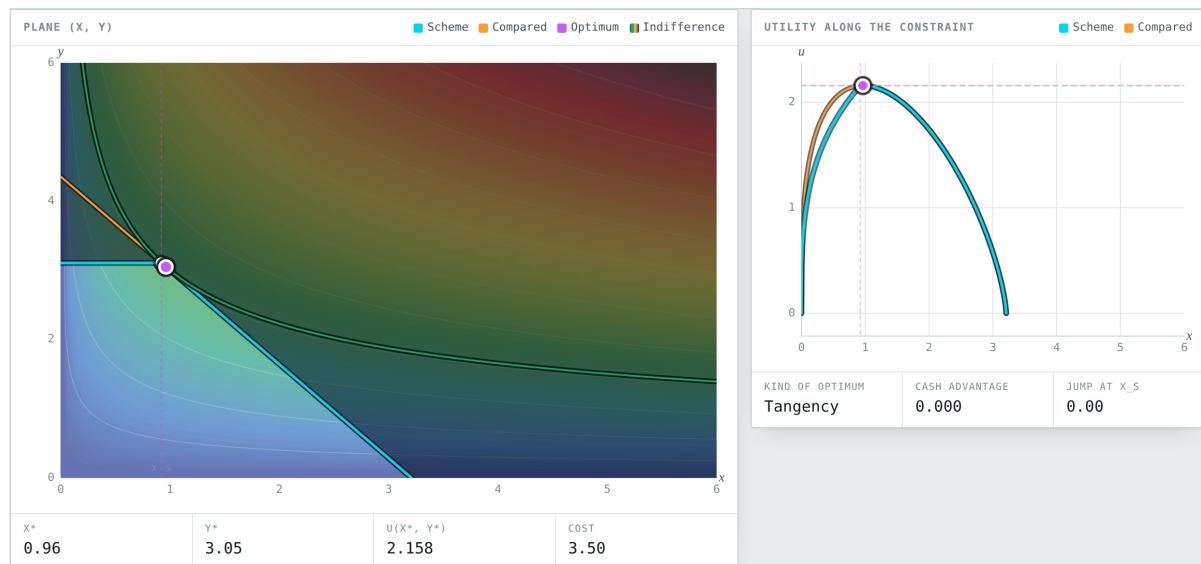


Figure 1. A non-exclusive in-kind subsidy. The first x_s units are free, and beyond that the market price applies: the frontier has a flat stretch and then a kink.

The point

The comparison that matters is between an in-kind subsidy and the cash transfer that costs the funder exactly the same. If the quota does not bind, the two tie. If it binds, cash wins, and the applet measures by how much.

1.1 Who it is for

The budget-constraint topic, and any public-economics class discussing in-kind transfers against cash. It assumes indifference curves and the idea of an optimum; it assumes no calculus.

2 Opening it

There is nothing to install. The whole application is one `index.html` file: it downloads no libraries, calls no server, and stores nothing outside your browser.

2.1 The three ways

Double-click the file. The quickest route. Find `nonlinear-budget/index.html` and open it. The browser shows it straight off the disk, with an address beginning `file://`. It works offline from

the first moment.

From the web. If someone has published it, the address ends in /nonlinear-budget/. Once the page has loaded you can disconnect: it needs the network for nothing else.

Served from a folder. For a class, putting the folder on any static server is enough. With Python installed:

```
python3 -m http.server 8000
```

then `http://localhost:8000/nonlinear-budget/` in the browser.

Note

Saving the page with `Ctrl + S` (or `Cmd + S` on a Mac) leaves a copy of your own that still works offline and that can travel on a memory stick or go out by email.

2.2 Language and theme

Two pairs of buttons sit at the top right.

Control	What it does
ES / EN	Switches the whole interface between Spanish and English on the spot. Nothing is recomputed, so you can switch mid-explanation without losing the state you had.
Dark / Light	Switches the theme. Dark is the original design; light exists for projectors and for printing.

Both choices are remembered in the browser, so next time it opens the way you left it.

2.3 Which browser

Any reasonably recent one: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone. On a narrow screen the panels stack one above the other instead of sitting side by side. No account, no permissions, no extensions.

3 The seven schemes

With price p_x , income m , quota x_s and subsidy rate s :

No. and name	Frontier
0 • Simple constraint	The usual line.
1 • In-kind subsidy, non-exclusive	Flat to x_s , then the market rate: a kink.
2 • Partial in-kind, non-exclusive	Price $(1 - s)p_x$ to x_s , then p_x : a shallower kink.
3 • Exclusive subsidy (state schools)	Take x_s free or buy at p_x : a vertical drop.
4 • Partial exclusive	A discount on the first x_s units, forfeited entirely if you exceed the quota.
5 • Cash equivalent (full)	Income $m + p_x x_s$, straight.
6 • Cash equivalent (partial)	Income $m + s p_x x_s$, straight.

Scheme 1 pairs with 5, and 2 with 6: the same cost to whoever funds the programme, with no

strings attached. Putting one against the other is what the **Compare with** control is for, and it is the point of the applet.

Note

With $s < 0$ the subsidy turns into a tax of rate $\tau = -s$ on the first x_s units. The slider reaches -1 , so the same seven schemes also serve for talking about selective taxes.

3.1 Kink versus notch

Schemes 1 and 2 produce a **kink**: the frontier changes slope but does not break. Schemes 3 and 4 produce a **notch**: exceeding the quota forfeits the whole subsidy, so the frontier falls vertically and there are bundles no amount of spending reaches.

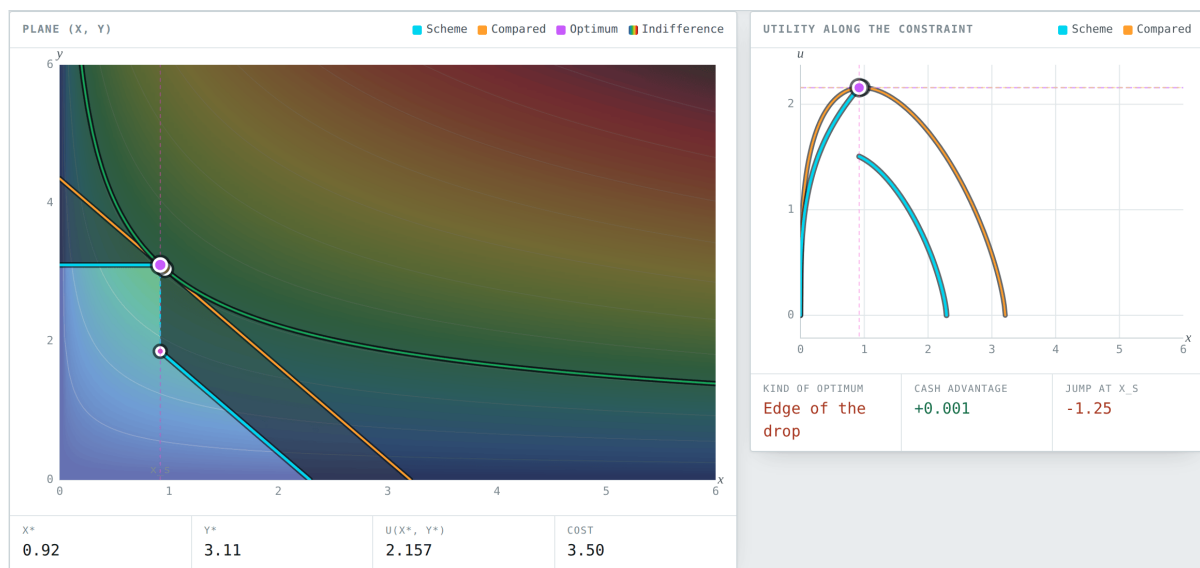


Figure 2. An exclusive subsidy. The frontier drops vertically at x_s : one more unit of x costs the whole quota.

4 The screen

Scheme. All seven, numbered 0 to 6.

Compare with. Draws a second scheme behind the first, and switches on the **Cash advantage** readout.

Preferences. Seven families, the last one writable.

Prices and income. p_x, p_y, m .

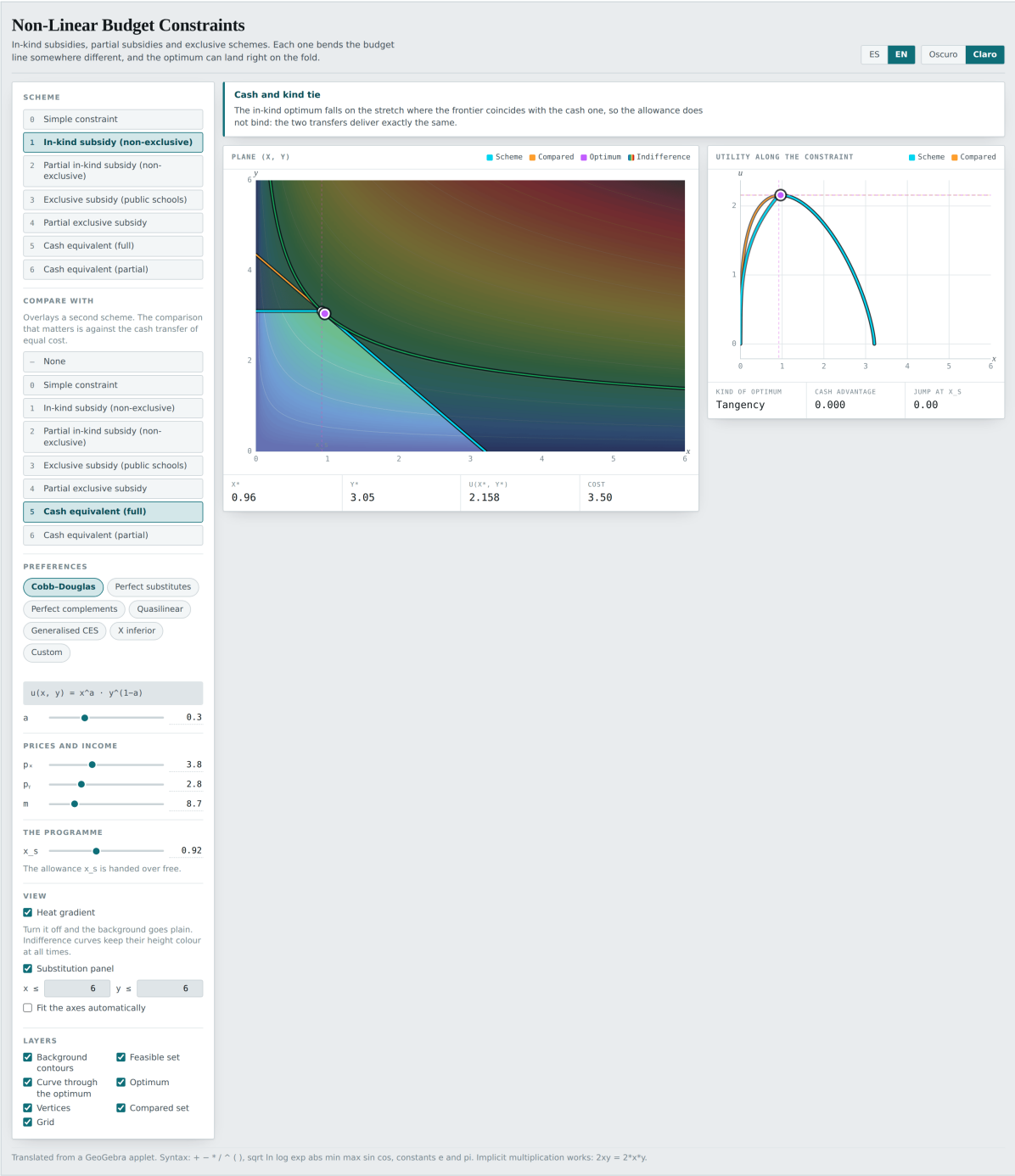
The programme. The quota x_s and the rate s .

View. Heat gradient, substitution panel, axes.

Layers. What gets drawn.

4.1 The two panels

Plane (x, y) . The utility map under a topographic ramp, everything out of reach veiled over, the chosen scheme's frontier, the compared one behind it, the indifference curve through the



optimum, and the vertices.

Utility along the constraint. $u(x, f(x))$ as x runs along the frontier. Its peak is the optimum, so “find the best affordable bundle” becomes “find the top of this curve”. On schemes 3 and 4 the curve visibly breaks in two: that gap is the forfeited subsidy.

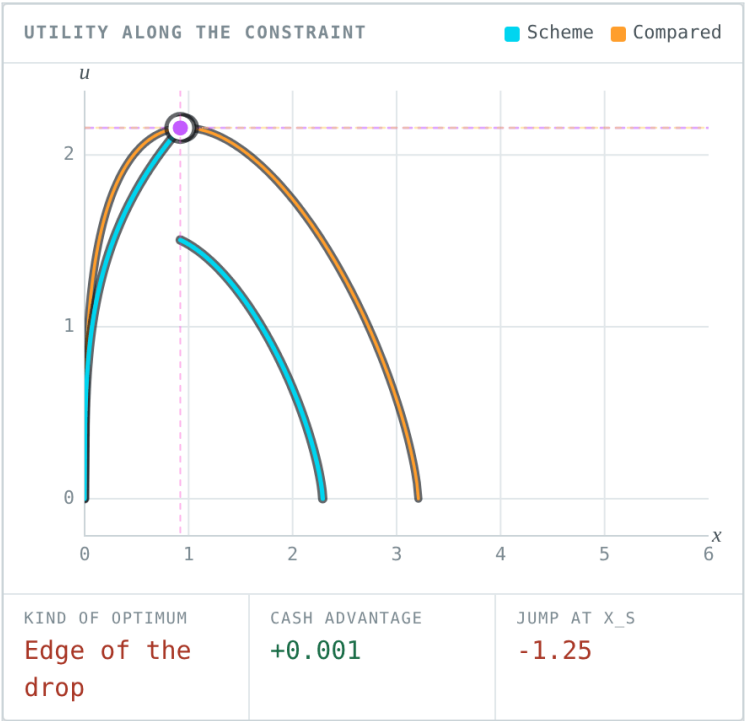


Figure 4. Utility along the constraint under an exclusive scheme. The gap between the two arcs is exactly the forfeited subsidy.

4.2 The dark theme

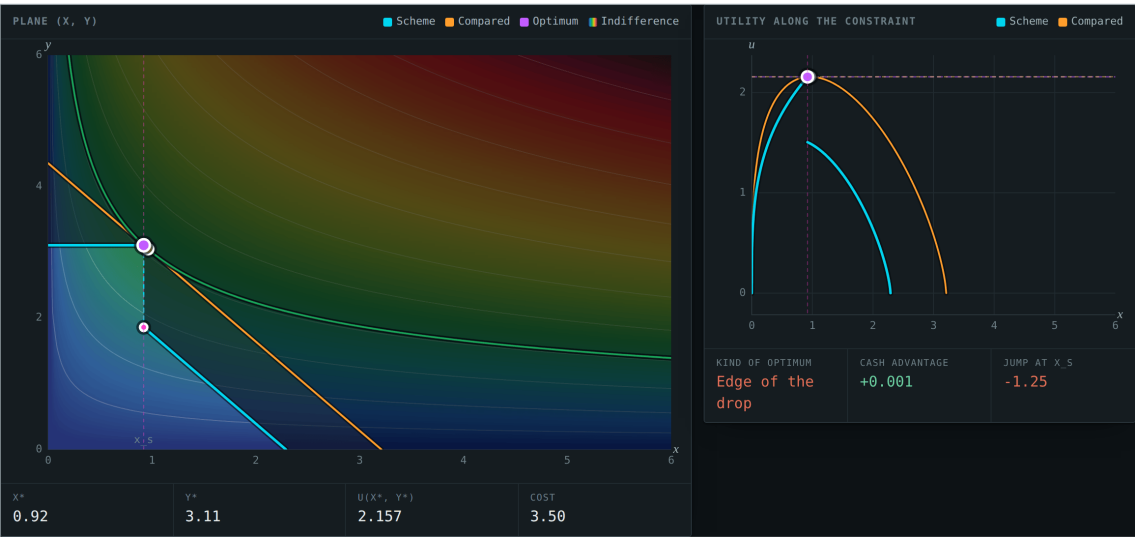


Figure 5. The plane and the substitution panel in the dark theme.

5 How the optimum is found

A first-order condition is the wrong instrument here. On these frontiers the best bundle is very often at a vertex, where the slope is undefined and no tangency holds; and on the exclusive schemes it sits at the right-hand end of the first piece with a strictly worse point immediately to its right.

1. Every piece of the frontier is scanned densely.
2. The best interior sample on each piece is refined by golden section.
3. *Every vertex is tested explicitly*, from both sides of every join.
4. Whichever wins is the optimum. Where it came from is what classifies it as a tangency, a kink, a corner, or the edge of a drop.

Kind of optimum	What it means
Tangency	Interior to a piece: the MRS equals the relative price prevailing there. The fold is not biting.
Kink	Exactly at the elbow. To the left the constraint is too flat and to the right too steep: best to stay at the fold. The consumer takes the quota exactly.
Corner	On an axis: it pays to spend everything on one good.
Edge of the drop	On an exclusive scheme, pinned at the edge. One more unit would leave the consumer strictly worse off.

6 The preferences

Family	Form
Cobb–Douglas	$x^a y^{1-a}$
Perfect substitutes	$ax + (1-a)y$
Perfect complements	$\min(ax, (1-a)y)$
Quasilinear	$8a \ln x + y$
Generalised CES	$\left[\frac{1}{2a} + \frac{1}{2(1-a)} \right]^{-1} (ax^\rho + (1-a)y^\rho)^{1/\rho}$
X inferior	$\ln(x - \underline{x}) - \frac{1}{1-\beta} \ln(\bar{y} - y)$
Custom	Whatever you type

One weight a runs through all of them: the exponent in Cobb–Douglas, the relative value in the linear and Leontief cases, the share in the CES.

Note

Two notes on the generalised CES. Its leading factor is written internally as $2a(1-a)$, which it equals identically: the literal $1/(2a) + 1/(2(1-a))$ divides by zero at both ends of the slider, while $2a(1-a)$ simply goes to zero there. Being a positive constant it rescales utility without touching preferences or the optimum. And at $\rho = 0$ the CES is undefined while its limit is Cobb–Douglas, so a slider resting on zero emits that limit rather than dividing by it.

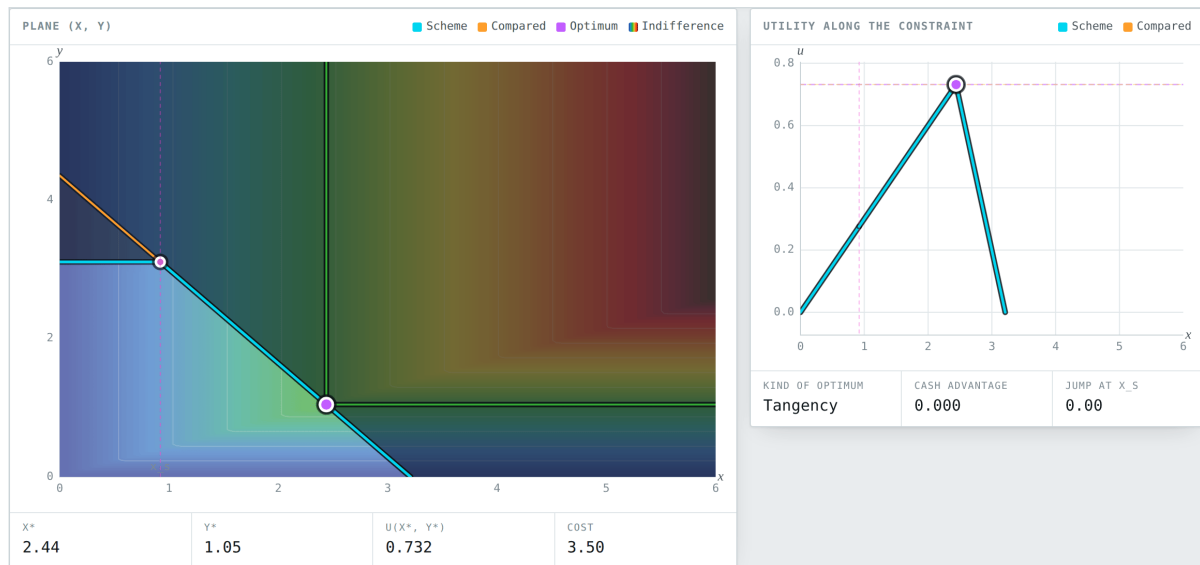


Figure 6. Perfect complements over an in-kind subsidy. With Leontief preferences the optimum is always at the elbow of the indifference curve, and here it competes with the elbow of the frontier as well.

7 The controls

Slider	What it is
p_x, p_y	Prices, bounded below at 0.2.
m	Income.
x_s	The programme's quota. It is clamped to what the income can actually buy, so the allowance never exceeds the budget.
s	The subsidy rate, -1 to 1 . Negative is a tax.
$\alpha, \rho, \beta, \underline{x}, \bar{y}$	Parameters of the chosen preference family; only the ones that family uses appear.

7.1 View

Heat gradient. Turn it off and the background goes plain. The indifference curves keep their height colour at all times: with the map showing it agrees with the ground underneath, and with the map off it is the only reading of height left.

Substitution panel. The utility-along-the-constraint reading is a supporting view, so it takes the smaller share of the width and can be dismissed entirely, giving the plane the full frame.

Axes. Fixed by default, set by the $x \leq$ and $y \leq$ boxes. An axis that refits on every slider move rennumbers as you drag, and then two settings cannot be compared by eye — which is the whole point of putting a scheme and its cash equivalent on one picture. Auto-fit stays available, and a warning appears if a frontier runs outside a fixed frame rather than being silently cropped.

7.2 Layers

Layer	What it draws
Background contours	Level curves of utility.
Feasible set	Veils what cannot be afforded.
Curve through the optimum	The indifference curve passing through it.
Optimum	The chosen point.
Vertices	The elbows and edges of the frontier.
Compared set	The frontier of the scheme being compared against.
Grid	The background rule.

8 Classroom activities

Activity 1 · The usual line

Settings: Scheme **0 · Simple constraint**, opening values.

What you should see: $x^* = 0.69$, $y^* = 2.18$, utility 1.539, cost 0.00. The kind of optimum is **Tangency**: the textbook case.

It is the baseline the next six schemes are read against. Write it on the board before going on.

Activity 2 · When the quota does not bite

Settings: Scheme **1 · In-kind subsidy, non-exclusive**, opening values. Set **Compare with to 5 · Cash equivalent (full)**.

What you should see: $x^* = 0.96$, $y^* = 3.05$, utility 2.158, cost 3.50. The **Cash advantage** reads 0.000 and the verdict says “Cash and in-kind tie”.

Look at why: the optimum lands on the stretch where both frontiers coincide. The quota $x_s = 0.92$ is smaller than what this consumer was going to buy anyway, so the programme hands over money and nothing else. An in-kind subsidy that does not constrain is a cash transfer.

Activity 3 · When the quota does bite

Settings: As above, but raise x_s to 2.20.

What you should see: $x^* = 2.20$ exactly — the whole quota, not one unit more — $y^* = 3.11$, utility 2.801, cost 8.36. The kind of optimum changes to **Kink** and the **Cash advantage** rises to +0.217.

At the same cost of 8.36, cash reaches a higher indifference curve. Ask why a government would then pick in-kind. The answers — paternalism, targeting, externalities from the subsidised good — are precisely the ones this model does not contain, which is why it is worth raising them here.

Activity 4 · A notch instead of a kink

Settings: Scheme **3 · Exclusive subsidy (state schools)**, opening values.

What you should see: The frontier falls vertically at x_s : the **Drop at x_s** readout gives −1.25. The

kind of optimum is **Edge of the drop**.

Switch on the **Substitution panel**. The utility-along-the-constraint curve breaks into two arcs that do not touch, and the gap between them is exactly the subsidy forfeited by crossing. Ask what happens to a family that needs barely one unit more than the quota.

Activity 5 · Tax instead of subsidy

Settings: Scheme **2 · Partial in-kind, non-exclusive**. Take s down to -0.6 .

What you should see: The fold reverses: the first x_s units now cost more, and the frontier is *concave* instead of convex. The optimum moves away from x .

It is a selective tax on the first few units, and the same picture explains it. Ask which real tax schemes look like this.

Activity 6 · Preferences that always end at the elbow

Settings: **Perfect complements** preferences, scheme **1 · In-kind subsidy**.

What you should see: With Leontief the optimum is always at the elbow of the indifference curve. Move x_s : as long as the consumer's elbow is to the right of the quota, the whole subsidy is taken up and then buying continues.

Compare with **Perfect substitutes**, where the optimum jumps from one axis to the other the moment the relative price crosses the marginal rate of substitution. The two extreme cases bracket the intuition.

Activity 7 · Fixed axes are not a whim

Settings: Any scheme with **Compare with** active. Tick and untick **Fit the axes automatically** while moving m .

What you should see: With automatic axes the numbers change at every step and the two frontiers appear to sit still. With fixed axes you can see which one moves and by how much. That is why the axes are fixed by default: comparison is the object of the applet, and a comparison whose scale moves on its own is not one.

Activity 8 · Build the case where in-kind wins

Settings: Free rein.

What you should see: You cannot, and that is the exercise. Whatever the preferences, cash of the same cost always reaches an indifference curve at least as high, because the feasible set under cash contains the one under in-kind.

The most that can be achieved is a tie, and the tie happens exactly when the quota does not bind. Letting the class try for five minutes is worth more than stating it.

9 Expression syntax

Element	What is allowed
Operators	+ - * / ^ ()
Functions	sqrt ln log log10 exp abs sin cos tan min max
Constants	e, pi
Implicit multiplication	2xy is $2 \cdot x \cdot y$
Associativity	^ binds to the right and tighter than unary minus: $-x^2$ is $-(x^2)$

10 Changes from the GeoGebra original

Discontinuities are modelled, not approximated. The original wrote schemes 3 and 4 as single If[...] functions, which GeoGebra draws with a connecting segment across the jump. Each scheme here is a list of separate pieces, so the drop stays a drop and the utility profile breaks where it should.

Colour range. $u_{\text{MIN}} := U(0.001, 0.001)$ is undefined for the Stone-Geary utility, whose domain needs $x > \underline{x}$; the ramp collapsed to a single colour for the whole “x inferior” example. The rewrite takes the 2nd percentile of the sampled field.

u_{MAX} used the same x-derived value for both coordinates, and exceeded $\bar{y}$ often enough to return NaN. Same fix.

Prices. p_X and p_Y sliders started at 0, making I/p_Y and the whole construction divide by zero. Both are bounded below by 0.2.

x_s and $\underline{x}$ are clamped to what the income can actually buy, so the allowance can never exceed the budget and the Stone-Geary utility stays defined.

Dead objects. $p_{X\text{low}}$ and $p_{X\text{high}}$ were computed from a formula that returns large negative numbers for every setting the sliders allow, and were displayed nowhere. Not carried over.

The cash comparison is computed, not left for the reader to eyeball: the cost of each scheme and the utility difference against the equal-cost cash transfer are readouts.

11 Troubleshooting

Symptom	What is happening
A warning that the frontier runs off	The axes are fixed and the drawing does not fit. Raise $x \leq$ or $y \leq$, or tick auto-fit.
The plane is plain	The Heat gradient is off. The curves keep their height colour regardless.
Cash advantage reads “—”	There is no scheme in Compare with .
The optimum ignores s	The quota is not binding: the optimum falls outside the subsidised stretch. Raise x_s .
Utility comes out NaN	In the “X inferior” family the domain needs $x > \underline{x}$ and $y < \bar{y}$. Adjust those two sliders.

12 Publishing it for a class

Any static host will do: GitHub Pages, Netlify, a university web directory, even a shared folder served over HTTP. The only thing to upload is the `index.html` file, inside a folder named however you want the address to read.

The repository ships a GitHub Actions workflow (`.github/workflows/pages.yml`) that publishes every applet at once on each push to the default branch. To switch it on, in the repository: *Settings* → *Pages* → *Source: GitHub Actions*. That is the one step the workflow cannot do for itself, because creating a Pages site needs administration rights that a `GITHUB_TOKEN` is never granted.

Note

To hand the applet to students without reliable internet, send them the `index.html` file directly. It weighs less than a photograph and opens with a double-click.

13 Licence and provenance

The application code is MIT. Use it, change it and pass it on, in class or out of it, with attribution.

This applet is a standalone rewrite of an original GeoGebra applet. The rewrite is not a literal translation: where the original had a construction error, a division by zero reachable with its own sliders, or a dead object inherited from whatever applet it was copied from, this version fixes it and says so. The section *Changes from the GeoGebra original* lists those changes one by one.

Everything is hand-written and dependency-free: the expression parser, the symbolic differentiation, the level-curve tracing and the drawing. In particular there is no `eval`, so what a student types into a text box cannot become code, and a strict content security policy does not break the page.